Dreamcast, the hackers choice

Selling for around $75 (approx Rs 3,700) with an Ethernet port, the Sega Dreamcast is currently the choice game console for information warfare as demonstrated at the Black Hat conference in Las Vegas held on July 31, 2002. As one example, the console has been converted into a bug that can be dropped into a corporate network with little or no detection.

Aaron Higbee of Foundstone and Chris Davis of RedSiren have developed a Linux toolkit that runs on the system, and when hooked up to a network, dials into the hacker's network. Known as the DC Phone Home kit, the software checks all available services such as HTTP, SMTP and SSH in a bid to make an outbound connection. It even checks for proxy servers hooked up to the network. Though the kit could be used on a range of hardware quite easily, the Dreamcast was chosen due to its innocuous nature. According to a report from SecurityFocus, during their presentation at Black Hat, when performing penetration tests, they often found themselves able to get physical access to the target's facilities for a few minutes.

As a defence against such attacks, Higbee and Davis suggested allowing only network connections from preassigned MAC addresses, along with the obvious tightening up on physical security. Other hardware that could be used in the phone home attack includes software for the Compaq iPAQ and a bootable x86 CD-ROM which can perform the attack using any available PC.

Def Con 10

Def Con, an annual gathering that holds the Alexis Park Hotel's all-time record for the most alcohol consumed by one group in a weekend, has always intended to be a place where the young can quietly talk tech. But on its 10th anniversary held from August 2 to 4, 2002, it became more of a party, with over 6,000 paid attendees flooding the premises—the three conference rooms used hold a combined total of just 2,800 people. San Francisco criminal defence attorney Jennifer Granick gave legal tips, while the MojoNation folks (an open source software for peer-to-peer sharing) spoke about their new better-than-Napster software. Arthur Money, assistant Secretary of Defense, spoke about hacking issues.

Toyota's driving companion

It smiles when you approach, opens its doors and rotates a seat towards you. It observes you in your house to study your music and TV preferences. The Toyota p.o.d (personalisation on demand) uses minipod, a portable terminal that constantly observes your lifestyle and habits, and adjusts the driving experience to provide greater comfort and satisfaction.

Introduced at the 2002 Tokyo Auto Show, the p.o.d creates a responsive, humanised environment that raises ergonomics to a new level. The car observes your behaviour and tailors the driving experience accordingly. More than an automobile, the p.o.d is an intuitive driving companion. Unfortunately, Toyota attaches a little note that reads, "Concept vehicle only. Not intended for production. Don't you wish."

snapshot

31.1 million

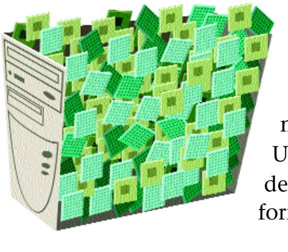
PCs were shipped in the second quarter of 2002

Source: *Bear Stearns*

Super duper computing

US supercomputers have long been considered the world's most powerful. But the Earth Simulator built in Japan with an old design runs five times faster than the previous record holder, a machine that simulates nuclear tests at Lawrence Livermore National Laboratory in the US. "This machine is so powerful that a researcher who uses it can do in one day what it takes a researcher in the US to do in one month," said Jack Dongarra, a University of Tennessee professor who tracks the world's 500 speediest computers.

Supercomputers have thousands of processors working in tandem to analyse the most complex issues, such as nuclear test simulations, aircraft designs and drug creation. The Earth Simulator is faster than all 15 of the biggest supercomputers in the US combined and will be used in climate and earthquake studies. It uses 5,104 processors to perform 35.9 trillion calculations a second. IBM, which built ASCI White, HP and other US supercomputer makers are working on even more powerful machines.



"We can do that in a heartbeat and with a lot less money," said Peter Ungaro, vice president of high performance computing at IBM. Earth Simulator costs about $350 million, while ASCI White cost $110 million.

Price isn't the only factor. Certain research problems work better on one supercomputer type over the other. Custom supercomputers, for instance, have bigger data pipes, which is critical in climate modelling and nuclear research, whereas this can be a drawback for off-the-shelf processors, which work best on data analysis in fields such as genetic research.

■ Creative to ship thumb-sized MP3 player ■ HP stops selling imaging products to Dell ■ .Net will link to Apache ■ Xbox to hit 6 new countries this year